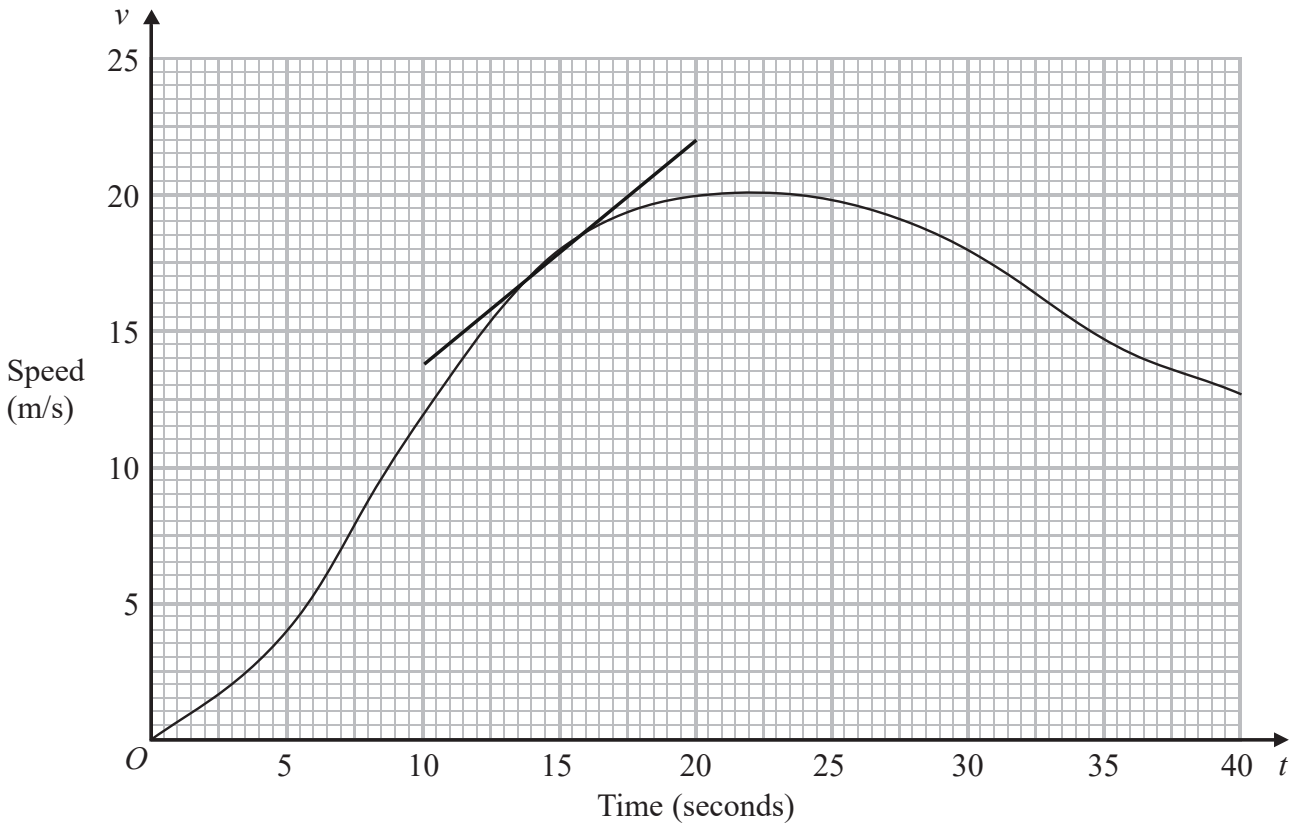
14 A car moves from rest.

The graph gives information about the speed, v metres per second, of the car t seconds after it starts to move.



(a) (i) Calculate an estimate of the gradient of the graph at $t = 15$

Tangent: rise ≈ 7.5 m/s, run ≈ 9 s

Gradient ≈ 7.5 ÷ 9 = 0.83 m/s²

1 mark
Tangent at t = 15.

1 mark
Rise ÷ run on tangent.

1 mark
Approx. 0.83 m/s².

0.83

(3)

(ii) Describe what your answer to part (i) represents.

This is the car's acceleration at 15 s.

1 mark
Rate of change of speed.

(1)

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(b) Work out an estimate for the distance the car travels in the first 20 seconds of its journey.
Use 4 strips of equal width.

Four strips of width 5 s: speeds 0, 4, 12, 18, 20 m/s.

Trapezia: 5/2(0+4) + 5/2(4+12)

+ 5/2(12+18) + 5/2(18+20)

= 10 + 40 + 75 + 95 = 220 m

1 mark
One correct 5 s trapezium.

1 mark
Four equal strips and sum.

1 mark
Distance ≈ 220 m.

220 m

(3)

(Total for Question 14 is 7 marks)

15 Make m the subject of the formula $f = \dfrac{3m + 4}{m - 1}$

(Total for Question 15 is 3 marks)

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